

Second-pass review: container-derivative uniqueness (U) and tangent-structure (E) notes

Rick

To: MacBeth (Kodamai)

2026-10-08

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Reviewed: *A uniqueness theorem for the container derivative* (U, 15 pp., “revised 7 October”) *The container derivative is a tangent structure* (E, 10 pp., “revised 7 October”)
Reviewed commit: 19e272d as stated in your email (UID 342); see note below
Previous reports: 2026-10-07 container review (UID 206); 2026-10-08 degree-two review (9844e32)
WIP commit: 7a6cb5d (grandpa-rick/work-in-progress)

Commit note. There is no commit 19e272d on scotmacbeth/work-in-progress. The attached PDFs are byte-identical (git blob hashes 35af605..., 5487b9c...) to papers/*.pdf at commit 19e372d8092f (“papers: apply Rick’s 2026-10-07 referee corrections to U + E”). I take that to be the intended commit and the hash in your email to be a typo. Neither PDF carries its commit hash on page 1. U’s title-page footnote cites only the old 57bac18.

Labels: FATAL, NEEDS FIX, COSMETIC. “l.” means the line in `pdftotext -layout` output. Section and lemma numbers are those of the 7-October PDFs.

Summary

No fatal errors. The module-level theorem holds: on free $\mathbb{N}$ -modules, the representable tangent structures $(-) \otimes D$ are exactly $D = \mathbb{N}$ and $D = W$, at every rank. I re-derived your new rank-free argument (footnote 1 of U) by hand. It is correct, and it is *better* than my flip suggestion. Lift-coassociativity alone forces ℓ_M to be the diagonal. You do not need to assume that c is the swap, and in fact the flip step is idle. Two inputs are left unstated, though, and the proof sits in a file that readers cannot see. My items (a) and (b) are fixed in substance. Some residues remain, and so does the old “only nontrivial tangent structure on polynomial functors” overclaim (U4 of my last report). Items (c) and (d) belong to a different note and are still open.

1 Status of the four earlier points

Point	Verdict	Where / residue
(a) Move Lemma 5 to $\text{Lin}(\text{SPoly})$	fixed in substance; residues	U §2, “The linear layer $\text{Lin}(\text{SPoly})$ and $\text{FreeMod}_{\mathbb{N}}$ ” (p. 5); Def. 2; §6 preamble (pp. 9–10, “all the maps in sight are morphisms of $\text{FreeMod}_{\mathbb{N}}$ ”). The augmentation is now a legitimate matrix morphism. Residues: Lemma 9’s proof still uses Poly hom-sets, and Lemma 7 still mixes objects and morphisms (N4).
(b) “Solid” $:= \ell_M$ iso	fixed	Def. 4 and Rem. 5 (p. 6). Residues: Thm 17 asserts that ℓ_M is an iso without proving it (N2), and Prop. 12 uses a different notion of solidity (N3).
(c) §7 claim refuted by $H^1(B\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$	open (not in these PDFs)	This point is about <i>Composition is a degree-two datum</i> (ae9d67a). You accepted it in UID 341, but 19e372d touches only U and E, and no revised degree-two PDF has arrived. A grep of U and E finds no H^1 , $B\mathbb{Z}/2$ or §7-program text. Please send that revision before the mini-course.
(d) Restore the Lanfranchi Thm 4.20 hypothesis	open (not in these PDFs)	Same note, also accepted in UID 341. Note that U’s “Examples 4.20–4.21” (Rem. 14, Table 1) refer to Lanfranchi–Lemay arXiv:2505.09080. That is a different paper from Lanfranchi arXiv:2609.03449, whose Thm 4.20 was misquoted. Nothing there needs to change.

2 The new claims

2.1 Rank-free uniqueness (Thm 13, footnote 1, Thm 21)

Write $\lambda: D \rightarrow D \otimes D$ for the map that induces $\ell = (-) \otimes \lambda$. On free modules every natural transformation $(-) \otimes D \Rightarrow (-) \otimes D \otimes D$ is of this form: evaluate at $\mathbb{N}$ and use naturality along the basis inclusions. Here is the chain, with my verdict on each step.

- (1) $\lambda(1) = 1 \otimes 1$ **and** $\lambda(M) \subseteq M \otimes M$. *Not stated in U*. It follows from $\ell \circ 0 = 0_T \circ 0$ together with $Tp \circ \ell = 0 \circ p$ and $p_T \circ \ell = 0 \circ p$. These say that $(\ell, 0)$ is a bundle morphism both ways. Then $(\text{id} \otimes \text{aug})\lambda(e_k) = 0$ and $(\text{aug} \otimes \text{id})\lambda(e_k) = 0$. Because $\mathbb{N}$ is zerosumfree, every coefficient on $1 \otimes 1$, $e_i \otimes 1$ and $1 \otimes e_j$ is 0.
- (2) $\lambda_M := \lambda|_M$ **is an isomorphism**. *Not proved in U*. Rem. 5 says that universality forces it “(§6)”. But §6 never constructs it, and Rem. 18 says explicitly that the proof “never constructs the solidity isomorphism”. The missing argument is short. Given (1), the canonical $v: T_2 \rightarrow V$ at $\mathbb{N}$ is

$$D_2 = \mathbb{N} \oplus M_1 \oplus M_2 \longrightarrow P' = \mathbb{N} \oplus M_{\text{out}} \oplus (M \otimes M), \quad (a, b, c) \mapsto a \cdot 1 \otimes 1 + c_{\text{out}} + \lambda_M(b).$$

This map is block-diagonal, so it is invertible iff λ_M is. That holds at every rank.

- (3) **Isos of free $\mathbb{N}$ -modules permute bases**. Correct at every cardinality, because the basis vectors are exactly the atoms. So $\lambda_M(e_k) = e_{\beta_1 k} \otimes e_{\beta_2 k}$ with $\beta = (\beta_1, \beta_2): K \rightarrow K \times K$ a bijection.
- (4) **Flip step**. $c\ell = \ell$ with λ_M onto $M \otimes M$ does force $c = \text{id}$ on $M \otimes M$. The step is correct but *idle*, since (5) never uses it. Present it as a side remark. The parenthesis “so c is not the swap when $\text{rank } M \geq 2$ ” is vacuous, because such M do not occur.

- (5) **Coassociativity.** Correct. $(\lambda \otimes \text{id})\lambda = (\text{id} \otimes \lambda)\lambda$ on e_k gives $\beta_1\beta_1 = \beta_1$, $\beta_2\beta_2 = \beta_2$ and $\beta_2\beta_1 = \beta_1\beta_2$. Each β_i is surjective because β is onto $K \times K$. A surjective idempotent is the identity, so β is the diagonal, which is onto only if $|K| \leq 1$. The orientation convention for ℓ_T versus $T\ell$ does not matter, since the equations are symmetric. At rank 1, $\lambda(\varepsilon) = \varepsilon \otimes \varepsilon$ satisfies both axioms.

Verdict. The argument is correct once (1) and (2) are added. As a write-up it is not yet a proof. Thm 13 is “not reproduced” and is cited to [12] (`proofs/2026-10-07-u8-flip-closes-f4.md`), but that file is not on GitHub at `19e372d` or at HEAD `b9e8158`. Thm 21’s proof also invokes Thm 13 *first*, even in the finite case. NEEDS FIX: turn footnote 1 into the proof of Thm 13, with (1) and (2) included and the flip moved to a remark. That is about 12 lines. Once this is done, the rank-free statement is proved in the note. The abstract, §1 and §9 should then say “lift-coassociativity” (plus universality), not “flip and coassociativity”. One more point: the footnote’s “independent $\triangleleft$ -monoidal argument, bypassing coassociativity” is unsupported. Either give it or delete it.

2.2 $\mathbb{N} \cdot y$: solid, but not a tangent structure

- *Not a tangent structure.* Confirmed by the argument above with $K = \mathbb{N}$. For the Cantor pairing β , the map β_1 is not the identity, since β is not the diagonal. So it violates coassociativity, and by (3)–(5) *every* bijection $\mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ does. This also holds for non-swap c .
- “Solid”. This is where the definitions equivocate (N3). Under Def. 4, solidity is a property of the vertical lift ℓ_M of a tangent structure. By Thm 13, $\mathbb{N} \cdot y$ underlies none, so under Def. 4 it is not solid, or rather the question is undefined. Prop. 12 proves something else: there is *an* isomorphism $M \cong M \otimes M$. That is exactly the abstract-iso notion that Def. 4 replaced. The true statement is: “ $\mathbb{N} \cdot y$ admits an iso $M \cong M \otimes M$ (abstractly solid), but no such iso is the vertical lift of a tangent structure.”

3 New and remaining issues

- N1.** NEEDS FIX The proof of Thm 13 is outsourced to an invisible file and is missing steps (1) and (2); see §2.1.
- N2.** NEEDS FIX Thm 17, last line (p. 11): “ M is solid ... ℓ_M is an isomorphism in each case”. This does not follow from $n \in \{0, 1\}$, because a module map $\mathbb{N} \rightarrow \mathbb{N}$ can be $\times 2$ or 0. Step (2) supplies the missing argument. Rem. 5’s forward reference to §6 should then point to it.
- N3.** NEEDS FIX Def. 4 vs Prop. 12, Rem. 14, the abstract and §9 (“ $\mathbb{N} \cdot y$ is solid”). Introduce “abstractly solid” ($\exists$ iso $M \cong M \otimes M$) separately from “the vertical lift is invertible”. Please also check Lanfranchi–Lemay’s own definition of “solid”. Table 1 claims the same notion, and I have not re-read their definition first-hand.
- N4.** NEEDS FIX Residues of (a).
- Lemma 9’s proof (p. 7) still says “morphisms between linear containers are just functions on shapes”. That is the Poly hom-set, not $\text{FreeMod}_{\mathbb{N}}$. Replace it with the atoms argument of (3). The statement itself is true.
 - Lemma 7 still identifies the object tensor $\mathbb{N}^S \otimes \mathbb{N}^T$ with $(S \cdot y) \triangleleft (T \cdot y)$. In $\text{Lin}(\text{SPoly})$, $S \cdot y$ is the 1×1 matrix $[[S]]$, so $\triangleleft$ gives the scalar $[[S][T]]$, while $M \otimes M$ is an object of rank $|S|^2$. The agreement is numerical across the object/morphism divide, which is the old U1 issue in

new clothes. §2 itself says “ $S \cdot y$ appears as a morphism”, but Def. 2 and Lemmas 9, 10 and 16 treat $M = S \cdot y$ as an object.

- None of the rank arithmetic needs Lemma 7, since $\text{rank}(M \otimes M) = n^2$ directly in modules. Demote Lemma 7 to a remark and do the counts in $\text{FreeMod}_{\mathbb{N}}$.

N5. NEEDS FIX The old U4 claim is unaddressed. On $\text{Lin}(\text{SPoly})$ the restricted combinator is $D[f] = f\pi_2$, by your own E Prop. 9. No one-hole content is visible there. These sentences still overclaim:

- Thm 1/21, last sentence (“the container derivative is the only nontrivial ...”);
- the end of §1 “Contribution”;
- §8 (“upgrades ... to ‘it is the only nontrivial one’”);
- §9 (“answers Cockett’s question in its strongest form: ... differentiation is the only nontrivial way”).

All of these were on my “do not say at Strathclyde” list. Say instead: “... exactly $\mathbb{N}$ and W ; the second is the restriction of the SPoly/∂ structure to linear maps.” The subtitle “on the finite free $\mathbb{N}$ -modules *in Poly*” is also wrong. These objects are not in Poly with Poly ’s morphisms; write $\text{Lin}(\text{SPoly})$ or $\text{FreeMod}_{\mathbb{N}}$.

N6. NEEDS FIX “The extensive category of $\mathbb{N}$ -modules” appears in the abstract, §1, §6, §9 and the Table 1 caption. It is false. A category with a zero object is extensive only if it is trivial, because $\mathcal{C}/(0 + 0) \simeq \mathcal{C}/0 \times \mathcal{C}/0$ fails. What you use is weaker and true: in free modules, the pullback of split coordinate maps is the coordinate submodule, so ranks add. That holds over any rig, *rings included*. So the “one new idea” is a remark about bookkeeping, not a mechanism that rings lack. Tone down “strictly weaker reasons than over a PID” and “isolating exactly which uses of ring structure were essential” in §8 accordingly.

N7. NEEDS FIX E’s headline still advertises the sketch. The abstract (“record a first restriction result: T is a strict $\triangleleft$ -monoidal base-change functor ... so the tangent bundle of a small category lives over $k[\varepsilon]/\varepsilon^2$ ”), §1 (“In §6 we record that it does”) and §7 (“Why it matters ... the headline is portable”) all contradict the sketch label of Prop. 11. My earlier E2 item asked for this to come out of the headline.

N8. NEEDS FIX (PROTOCOL) Neither PDF has its commit hash on page 1, and the hash in the email is wrong (19e272d vs 19e372d). Commit the tex, insert the hash, recompile, then commit the PDF.

Cosmetic.

C1. U §2 (p.5, l.211) still says “We work with *finitary* containers”. E1 was fixed in E but not in U.

C2. $\text{FreeMod}_{\mathbb{N}}$ means “finite free” in §1 (l.69) and on p.7 (“objects are the finite free $\mathbb{N}$ -modules”), but “all free” in §2 (l.223). Thm 21 needs “all free”.

C3. Rem. 18: “*In Poly* that fibre is a direct coproduct” (old S4 wording).

C4. Rem. 19: “solidity coincides with indecomposability” is still unsupported.

C5. Lemma 15 is still a Poly lemma, used for $\text{FreeMod}_{\mathbb{N}}$ pullbacks. One sentence suffices instead: “ $A \otimes -$ with A free preserves split coordinate pullbacks”.

- C6.** §1 (1.110): Lanfranchi–Lemay work on the opposite of commutative *algebras*, not “the opposite of the category of modules”. This is unchanged from my last report.
- C7.** Rem. 22: $c\ell = \ell$ pins c only on $M \otimes M$. On $1 \otimes \varepsilon$ and $\varepsilon \otimes 1$ you also need $Tp \circ c = p_T$. Add one line.
- C8.** E: there is still no sentence relating E to U (“E’s $T(n) = 2n$ restricted to linear maps is U’s $(-) \otimes W$ ”), and “every object is linear, so the question says nothing” is unchanged.

Not checked by me: the Lean claims, and CC2014 Prop. 4.7 (you report verifying it).

4 Recommended registry grades

Claim	Grade	Note
U Thm 21, module form $((-) \otimes D$ on free $\mathbb{N}$ -modules: exactly $\mathbb{N}$, W ; every rank)	checked-sober	becomes <i>proved</i> once N1 and N2 are inlined
U Thm 11/17 (finite rank)	checked-sober	gap N2 is one paragraph
“ ∂ is the only nontrivial tangent structure”	sketched (over-claim)	true only as “restriction to linear maps” (N5)
$\mathbb{N} \cdot y$ not a tangent structure	checked-sober	coassociativity alone
$\mathbb{N} \cdot y$ “solid”	definitional	needs N3
E Thm 4 (finite-support SPoly is a CDC/tangent category)	checked-sober	fix E1 is done
E Prop. 11 (directed containers)	sketched	labelled correctly; headline is not (N7)

Overall: checked-sober, not peer-reviewed or proved, until N1–N3 and N5 are in the text. All of these are short write-up fixes, and none needs new mathematics.

Safe to say at Strathclyde (13 Oct)

“On free $\mathbb{N}$ -modules, of any rank, the representable first-order tangent structures $(-) \otimes D$ are exactly the trivial one and the dual numbers W . The proof works by rank rigidity at finite rank, and by lift-coassociativity, which forces the vertical lift to be the diagonal, at every rank. W is the restriction of the container-derivative tangent structure on SPoly to its linear maps.” Do not say “ $\mathbb{N} \cdot y$ is solid” without the qualifier “abstractly”, and do not say that ∂ is unique among tangent structures on polynomial functors.